1. Introduction

Temperature sensor interfacing is the process of connecting a temperature sensor to a microcontroller or electronic circuit to measure and monitor temperature.

A temperature sensor detects changes in temperature and converts them into an electrical signal. This signal is processed by a microcontroller, such as Arduino, to display the temperature or control devices like fans and alarms.

A common example is the **LM35 temperature sensor**, which produces an output voltage proportional to the temperature in degrees Celsius.

2. Required Components



1. LM35 Temperature Sensor

Detects temperature and produces an analog output voltage.



2. Arduino Uno

Reads and processes the sensor's output voltage.



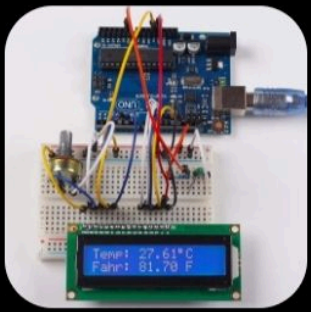
3. Breadboard and Jumper Wires

Connect the components and establish electrical connections.



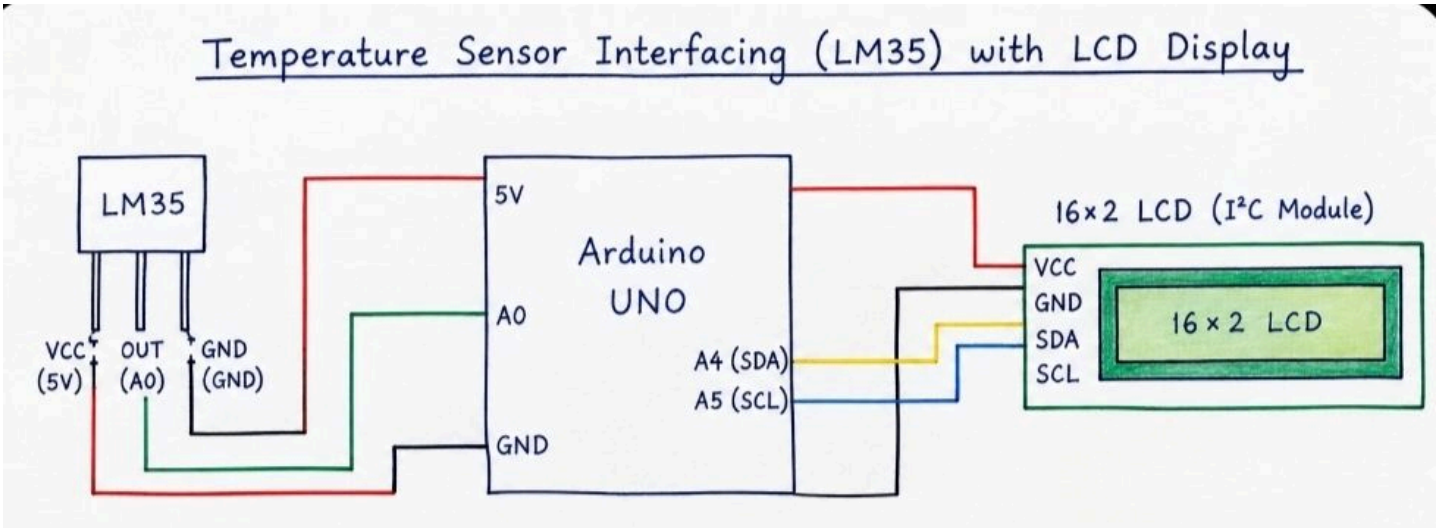
4. Power Supply / USB Cable

Supplies power to the Arduino and sensor.



5. LCD Display (Optional)

Displays the measured temperature. A computer's Serial Monitor can be used instead.


Temperature Sensor Interfacing (LM35) with LCD Display
LM35
VCC (5V)
OUT (A0)
GND (GND)
5V
A0
GND
Arduino UNO
A4 (SDA)
A5 (SCL)
16×2 LCD (I²C Module)
VCC
GND
SDA
SCL
16 × 2 LCD

5. Working Procedure — Step by Step

1. **Connect the sensor:** Connect the LM35 power pin to 5V, output pin to A0, and ground pin to GND on the Arduino.
2. **Power on the circuit:** Connect the Arduino to a suitable power supply using a USB cable.
3. **Read the output:** The LM35 generates an analog voltage according to the surrounding temperature.
4. **Convert the signal:** The Arduino reads the voltage and converts it into degrees Celsius.
5. **Display the result:** The measured temperature is displayed on the Serial Monitor or LCD.
6. **Monitor changes:** When the surrounding temperature changes, the sensor output changes, and the displayed reading updates.

7. Advantages

- Simple and easy to interface with microcontrollers.
- Provides temperature readings directly in Celsius.
- Low cost and low power consumption.
- Suitable for continuous temperature monitoring.
- Does not require complex signal conditioning for basic applications.

8. Disadvantages

- The LM35 provides analog output, which can be affected by electrical noise.
- Measurement accuracy depends on the sensor and the analog-to-digital converter.
- It has a limited operating temperature range.
- It requires suitable wiring and power connections.
- It may not be ideal for applications requiring very high precision without calibration.

9. Applications



Home temperature monitoring

Measures indoor temperature and supports automatic fan control.



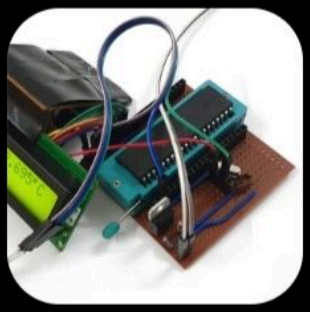
Industrial automation

Monitors machinery and equipment temperature.



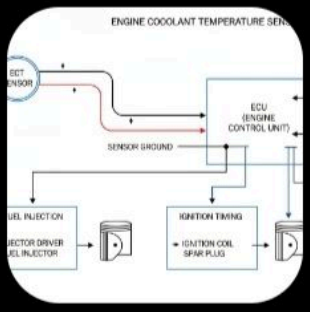
Agriculture

Monitors greenhouse and plant-growing environments.



Electronic devices

Helps monitor temperature in embedded systems and electronic equipment.



Automotive systems

Supports temperature monitoring in suitable vehicle applications.

10. Conclusion

Temperature sensor interfacing is an important application of embedded systems. By connecting an LM35 temperature sensor to an Arduino, temperature can be measured, processed, and displayed in real time. This system is simple, economical, and useful in domestic, industrial, agricultural, and electronic applications.